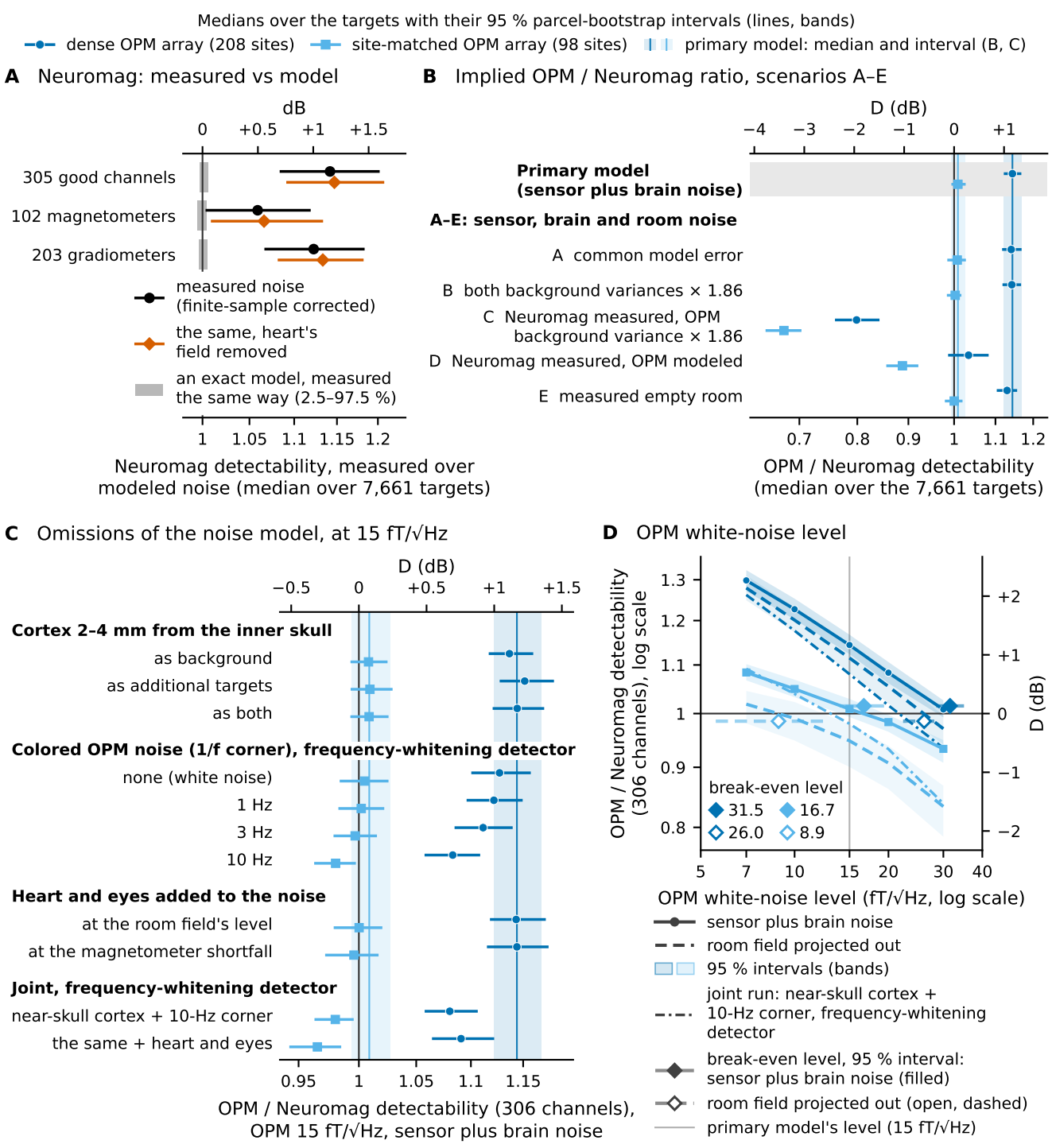
Medians over the targets with their 95 % parcel-bootstrap intervals (lines, bands)
dense OPM array (208 sites)
site-matched OPM array (98 sites)
primary model: median and interval (B, C)
A Neuromag: measured vs model
dB
0
+0.5
+1
+1.5
305 good channels
102 magnetometers
203 gradiometers
measured noise (finite-sample corrected)
the same, heart's field removed
an exact model, measured the same way (2.5–97.5 %)
1
1.05
1.1
1.15
1.2
Neuromag detectability, measured over modeled noise (median over 7,661 targets)
B Implied OPM / Neuromag ratio, scenarios A–E
D (dB)
−4
−3
−2
−1
0
+1
Primary model (sensor plus brain noise)
A–E: sensor, brain and room noise
A common model error
B both background variances × 1.86
C Neuromag measured, OPM background variance × 1.86
D Neuromag measured, OPM modeled
E measured empty room
0.7
0.8
0.9
1
1.1
1.2
OPM / Neuromag detectability (median over the 7,661 targets)
C Omissions of the noise model, at 15 fT/√Hz
D (dB)
−0.5
0
+0.5
+1
+1.5
Cortex 2–4 mm from the inner skull
as background
as additional targets
as both
Colored OPM noise (1/f corner), frequency-whitening detector
none (white noise)
1 Hz
3 Hz
10 Hz
Heart and eyes added to the noise
at the room field's level
at the magnetometer shortfall
Joint, frequency-whitening detector
near-skull cortex + 10-Hz corner
the same + heart and eyes
0.95
1
1.05
1.1
1.15
OPM / Neuromag detectability (306 channels), OPM 15 fT/√Hz, sensor plus brain noise
D OPM white-noise level
OPM / Neuromag detectability (306 channels), log scale
1.3
1.2
1.1
1
0.9
0.8
+2
+1
0
−1
−2
D (dB)
break-even level
31.5
16.7
26.0
8.9
5
7
10
15
20
30
40
OPM white-noise level (fT/√Hz, log scale)
sensor plus brain noise
room field projected out
95 % intervals (bands)
joint run: near-skull cortex + 10-Hz corner, frequency-whitening detector
break-even level, 95 % interval: sensor plus brain noise (filled)
room field projected out (open, dashed)
primary model's level (15 fT/√Hz)